

Ethanol Blending in Gasoline – Luxembourg

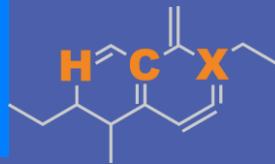
September 2025



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FARO90

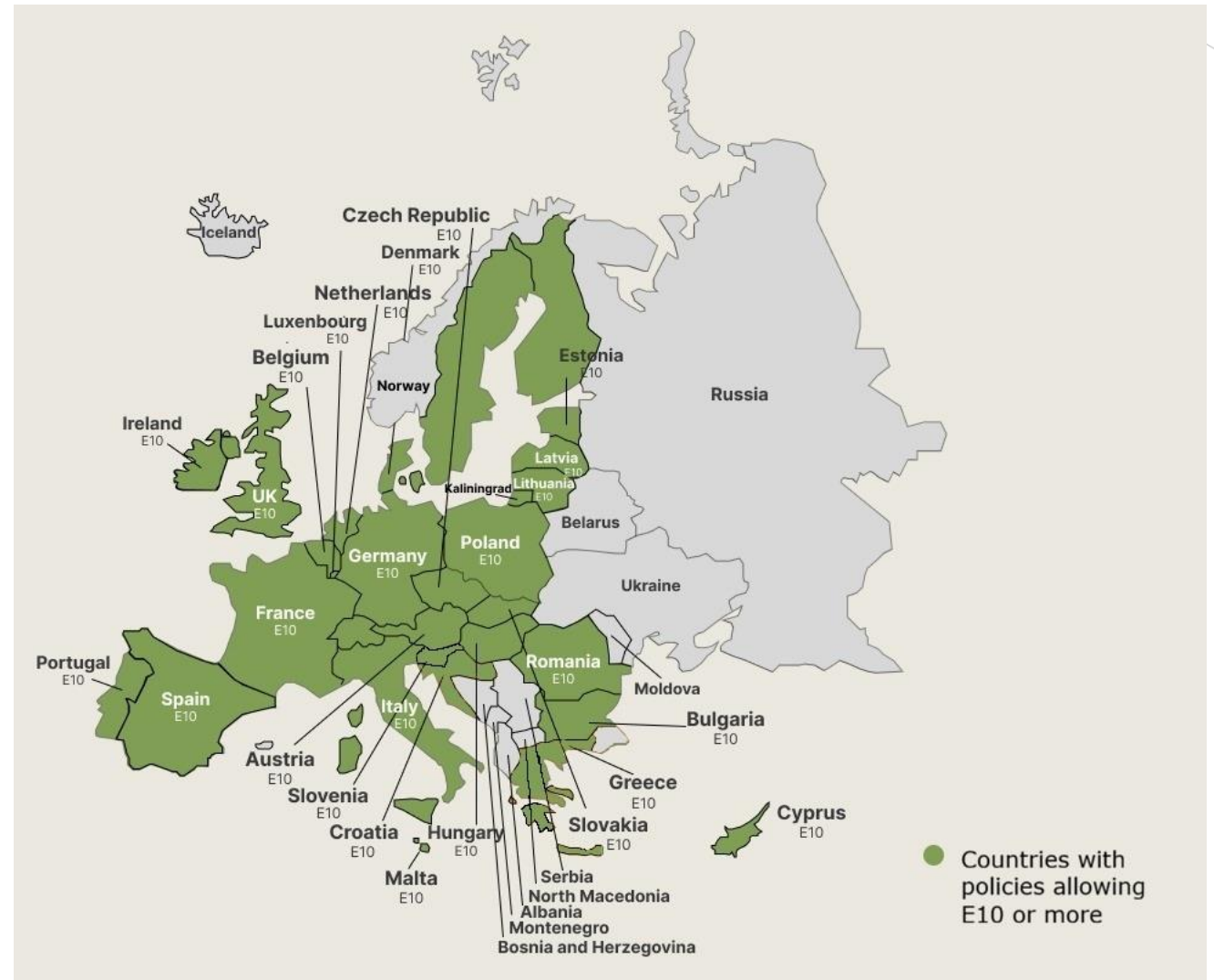


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Ethanol Blending in Europe

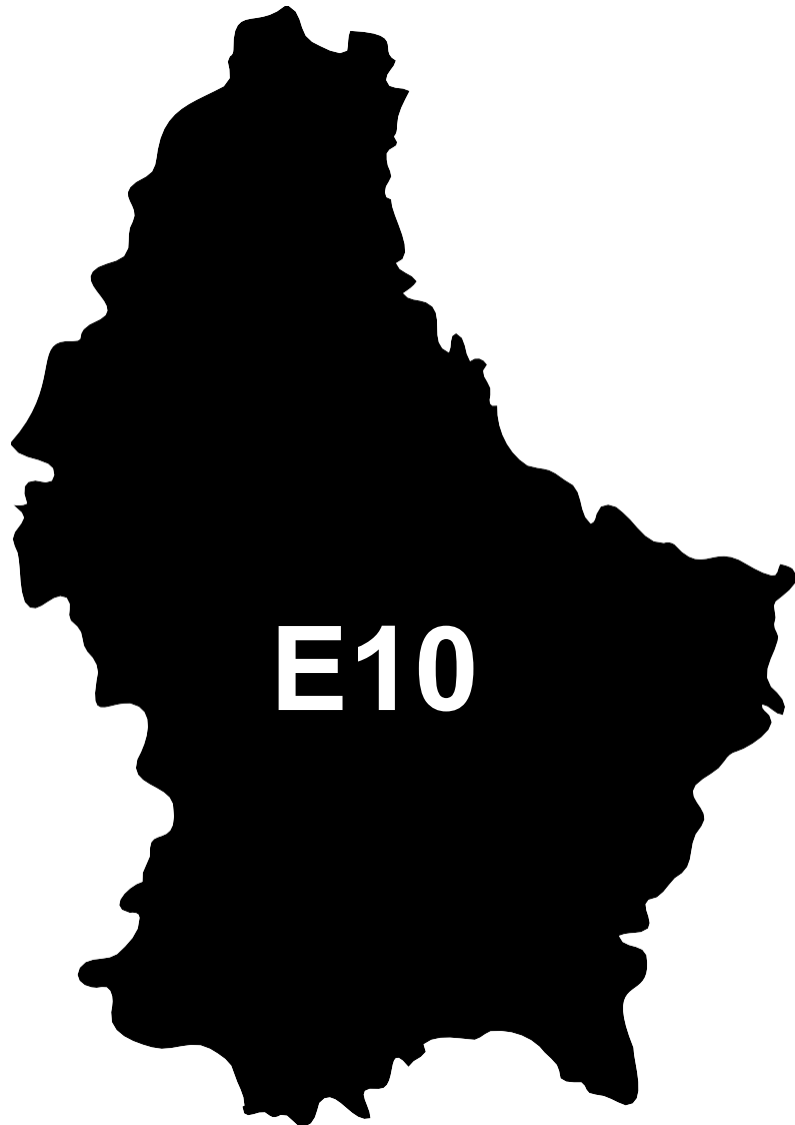
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Luxembourg

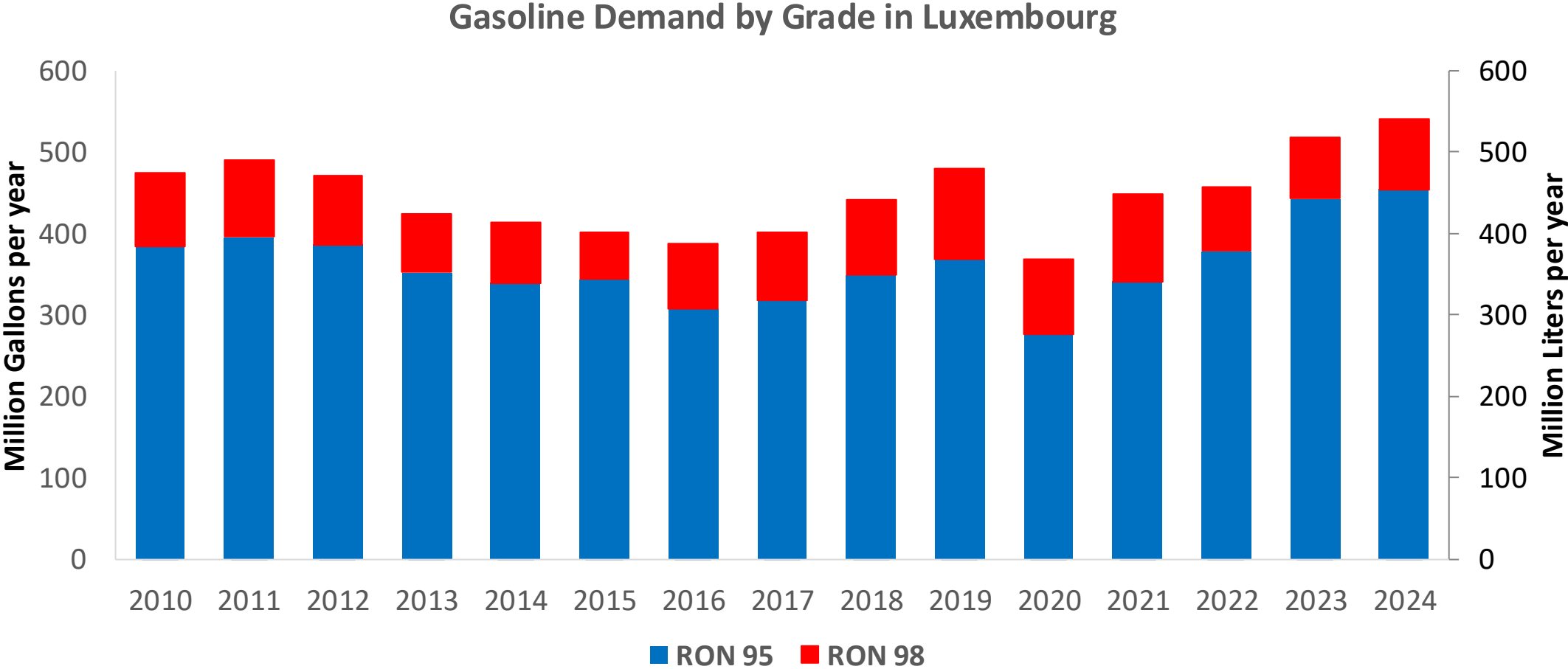


In 2024, gasoline consumption in Luxembourg was 540 million liters (143 million gallons) and growth rate was 2.4%. There is no gasoline production. Gasoline net imports were 540 million liters (143 million gallons) correspondingly. Luxembourg complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.5 RON.

In 2024, ethanol consumption in Luxembourg was 29 million liters (8 million gallons) representing 5.3% of gasoline demand. Net Imports were 29 million liters (8 million gallons). Ethanol sources are Sugar Cane 80% and Corn 20%.

Source: Eurostat, European Commission

Gasoline Demand in Luxembourg



Source: Eurostat

The FARO90 logo is shown next to a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

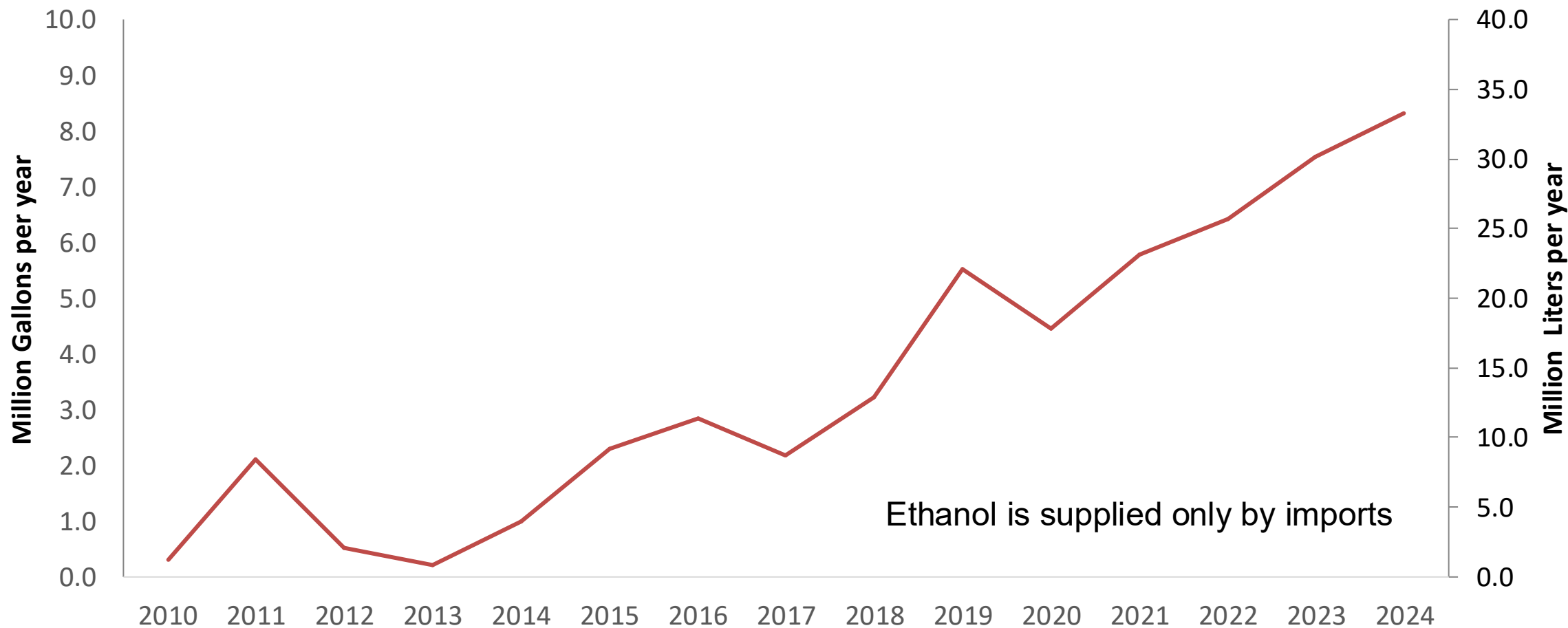
This stacked bar chart illustrates the annual consumption of RON 95 and RON 98 gasoline in the United States from 2010 to 2024. The left vertical axis measures consumption in Million Gallons per year (0 to 180), while the right vertical axis measures it in Million Liters per year (0 to 700). The horizontal axis lists the years from 2010 to 2024. Each bar is composed of two segments: a blue segment representing RON 95 and a red segment representing RON 98. The total height of each bar represents the combined consumption of both grades. The data shows a general upward trend in total consumption over the period, with a notable dip in 2020. The proportion of RON 95 consumption has increased significantly in the later years, particularly in 2024.

Year	RON 95 (Million Gallons)	RON 98 (Million Gallons)	Total (Million Gallons)
2010	102	48	150
2011	105	50	155
2012	103	44	147
2013	93	39	132
2014	90	39	129
2015	91	31	122
2016	81	43	124
2017	84	44	128
2018	92	49	141
2019	98	59	157
2020	73	49	122
2021	90	56	146
2022	101	41	142
2023	118	40	158
2024	120	45	165

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Ethanol Balance in Luxembourg

Ethanol Demand in Luxembourg



Ethanol is supplied only by imports

Source: Eurostat

Gasoline Quality in Luxembourg

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.7							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a double bond between carbons 1 and 2, with substituents at various positions, including a methyl group at C3, a propyl group at C4, and a vinyl group at C5. The carbons are labeled with orange letters H, C, and X.

Catalytic Gasoline
Light Primary Naphta
Heavy Primary Naphta
Reformate
Isopentane
Alkylate
Isomerate
Hydrocracked Gasoline
Coker Naphta
n-Butane
MTBE
ETBE
TAME
Aromatics

Country	Threat Category 1 (Dark Blue)	Threat Category 2 (Red)	Threat Category 3 (Grey)	Threat Category 4 (Purple)	Threat Category 5 (Light Grey)	Threat Category 6 (Orange)	Threat Category 7 (Yellow)	Threat Category 8 (Brown)	Threat Category 9 (Dark Blue)	Threat Category 10 (Light Yellow)	Threat Category 11 (Dark Purple)	Threat Category 12 (Dark Blue)
Austria	15%	15%	15%	15%	10%	10%	10%	10%	5%	5%	5%	5%
Belgium	18%	18%	18%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Bulgaria	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Croatia	15%	15%	15%	20%	5%	5%	5%	15%	10%	5%	5%	5%
Cyprus	15%	15%	15%	20%	10%	10%	10%	10%	5%	5%	5%	5%
Czechia	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Denmark	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Estonia	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Finland	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
France	20%	20%	20%	18%	5%	5%	5%	10%	5%	5%	5%	5%
Germany	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Greece	15%	15%	15%	20%	10%	10%	10%	10%	5%	5%	5%	5%
Hungary	12%	18%	15%	18%	5%	5%	5%	15%	10%	5%	5%	5%
Ireland	5%	15%	15%	25%	5%	15%	10%	5%	15%	5%	5%	5%
Italy	15%	15%	15%	20%	5%	5%	5%	15%	10%	5%	5%	5%
Latvia	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Lithuania	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Luxembourg	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Malta	15%	15%	15%	20%	5%	5%	5%	15%	10%	5%	5%	5%
Netherlands	18%	18%	18%	18%	5%	5%	5%	15%	5%	5%	5%	5%
Poland	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Portugal	15%	15%	15%	18%	5%	5%	5%	10%	5%	5%	5%	5%
Romania	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Slovakia	10%	5%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Slovenia	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Spain	15%	15%	15%	20%	5%	5%	5%	15%	5%	5%	5%	5%
Sweden	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
UK	5%	15%	15%	25%	5%	15%	10%	5%	15%	5%	5%	5%

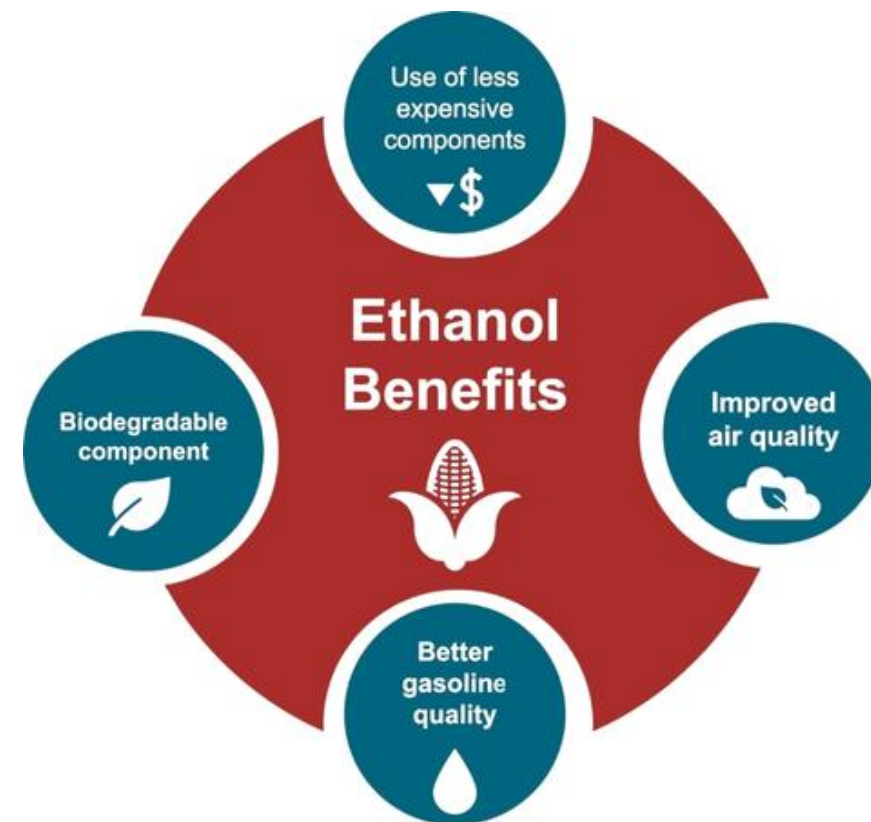
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

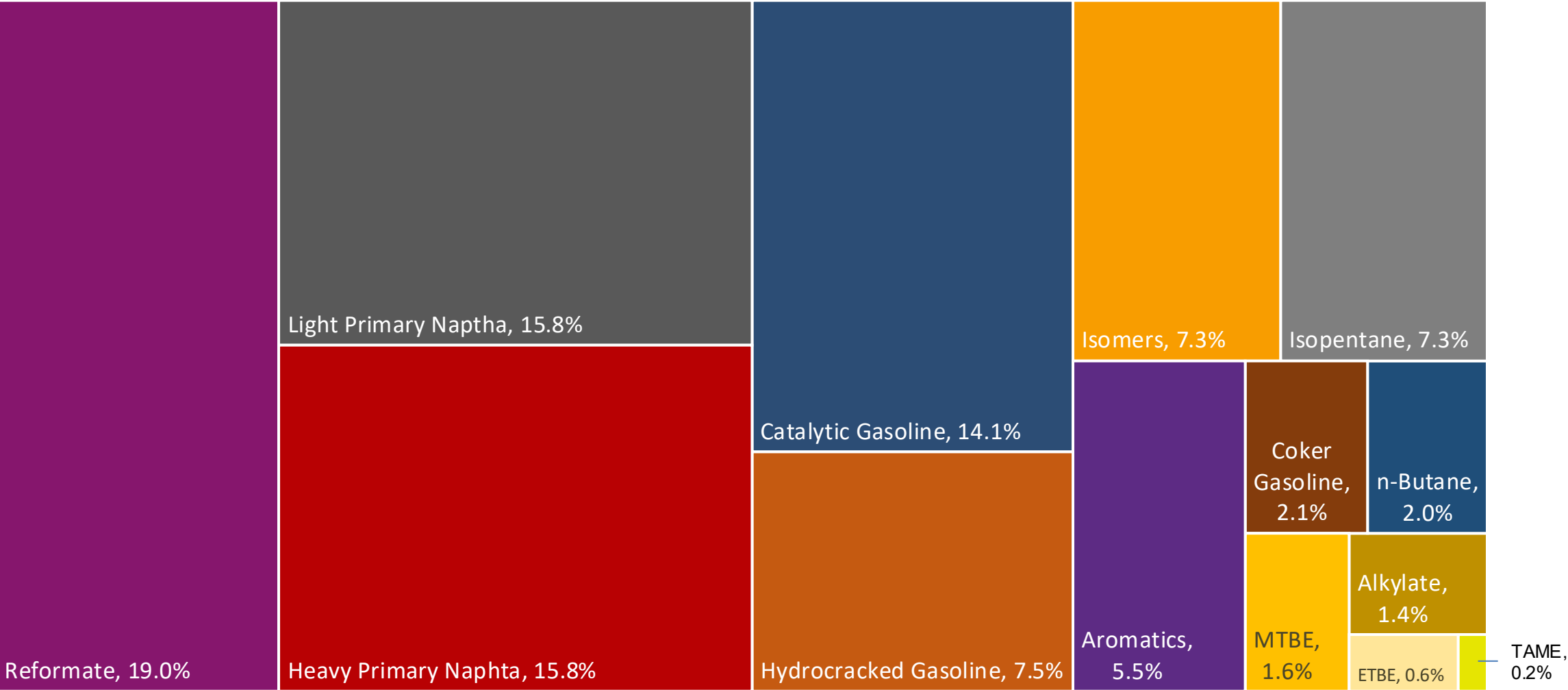
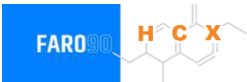
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

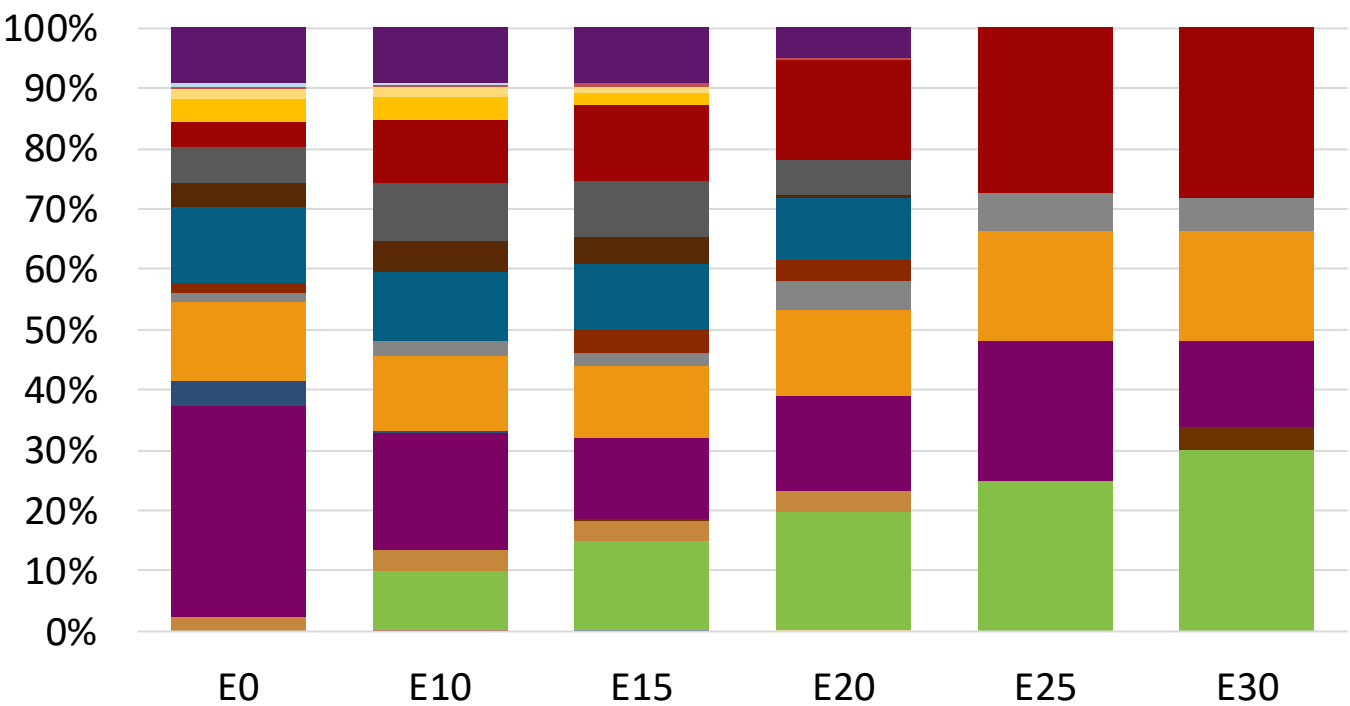
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Luxembourg Available Blending Components



Luxembourg – Gasoline 95 – Constant Octane

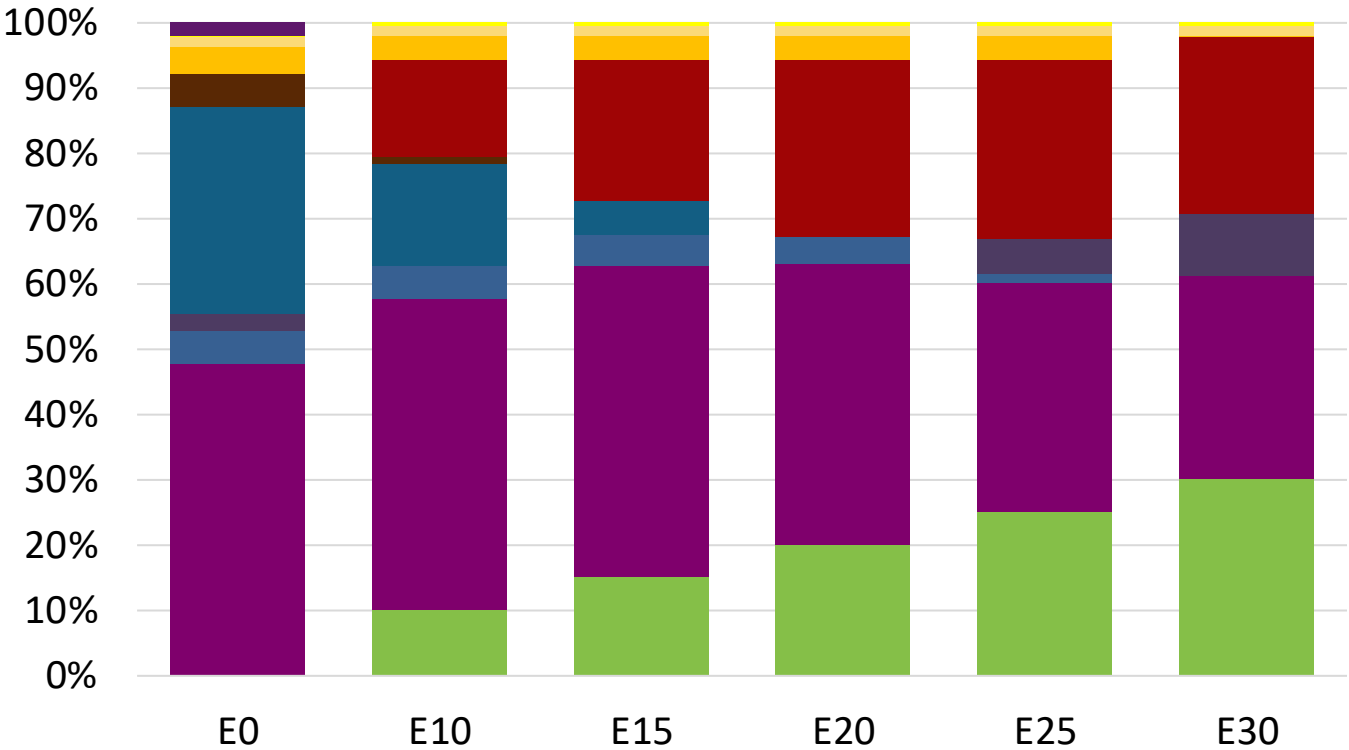


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

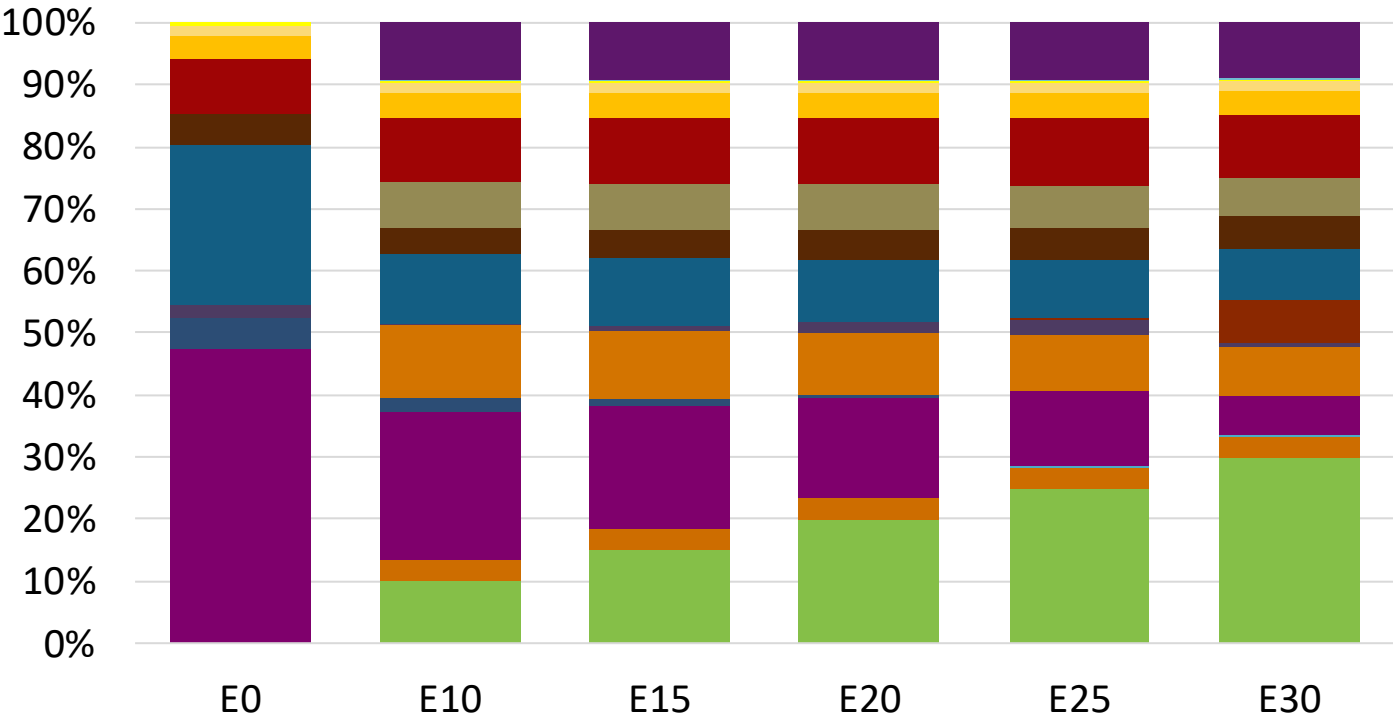
Luxembourg - Gasoline 98 - Constant Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

Luxembourg – Gasoline 95 – Increased Octane

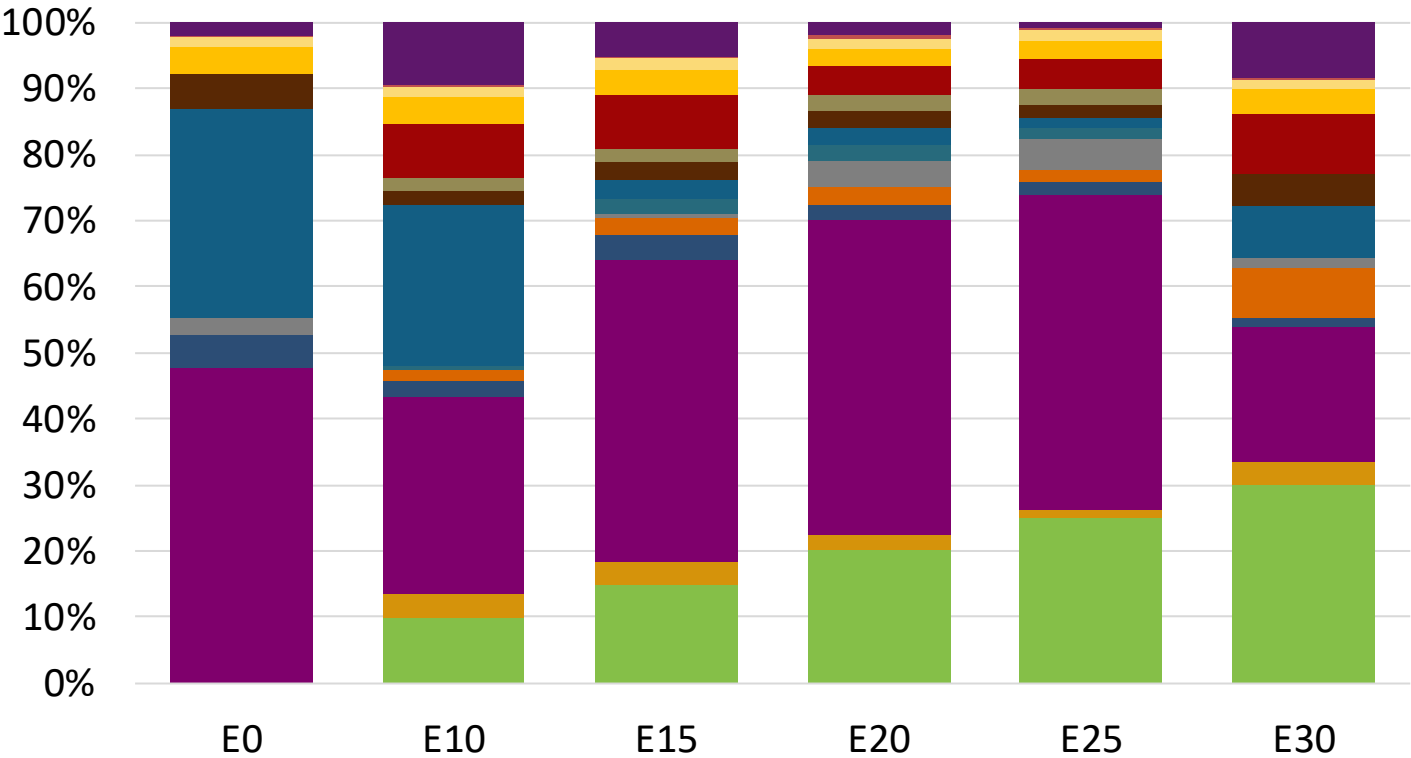


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Source: Faro90

Luxembourg – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)		98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$	2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

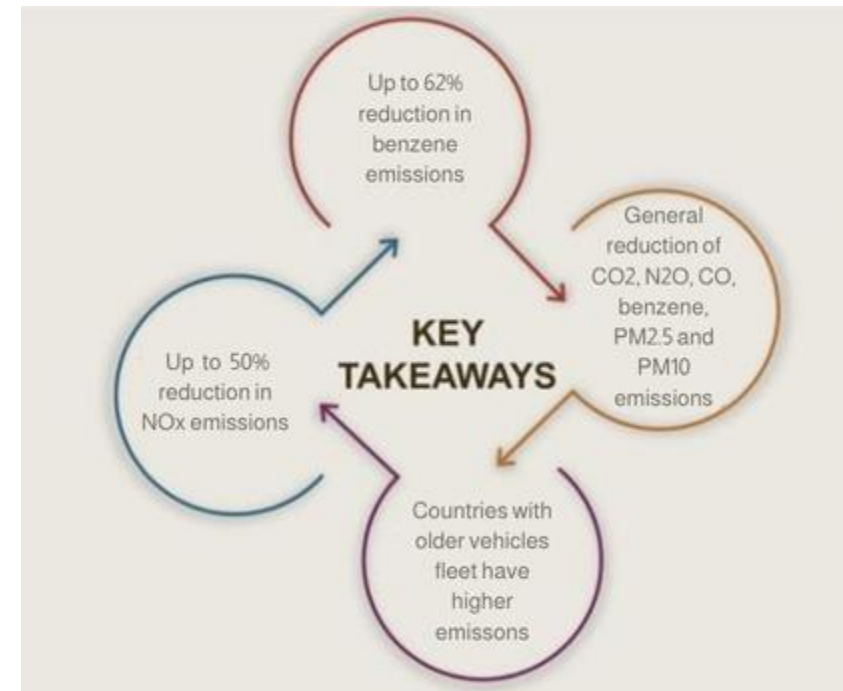
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

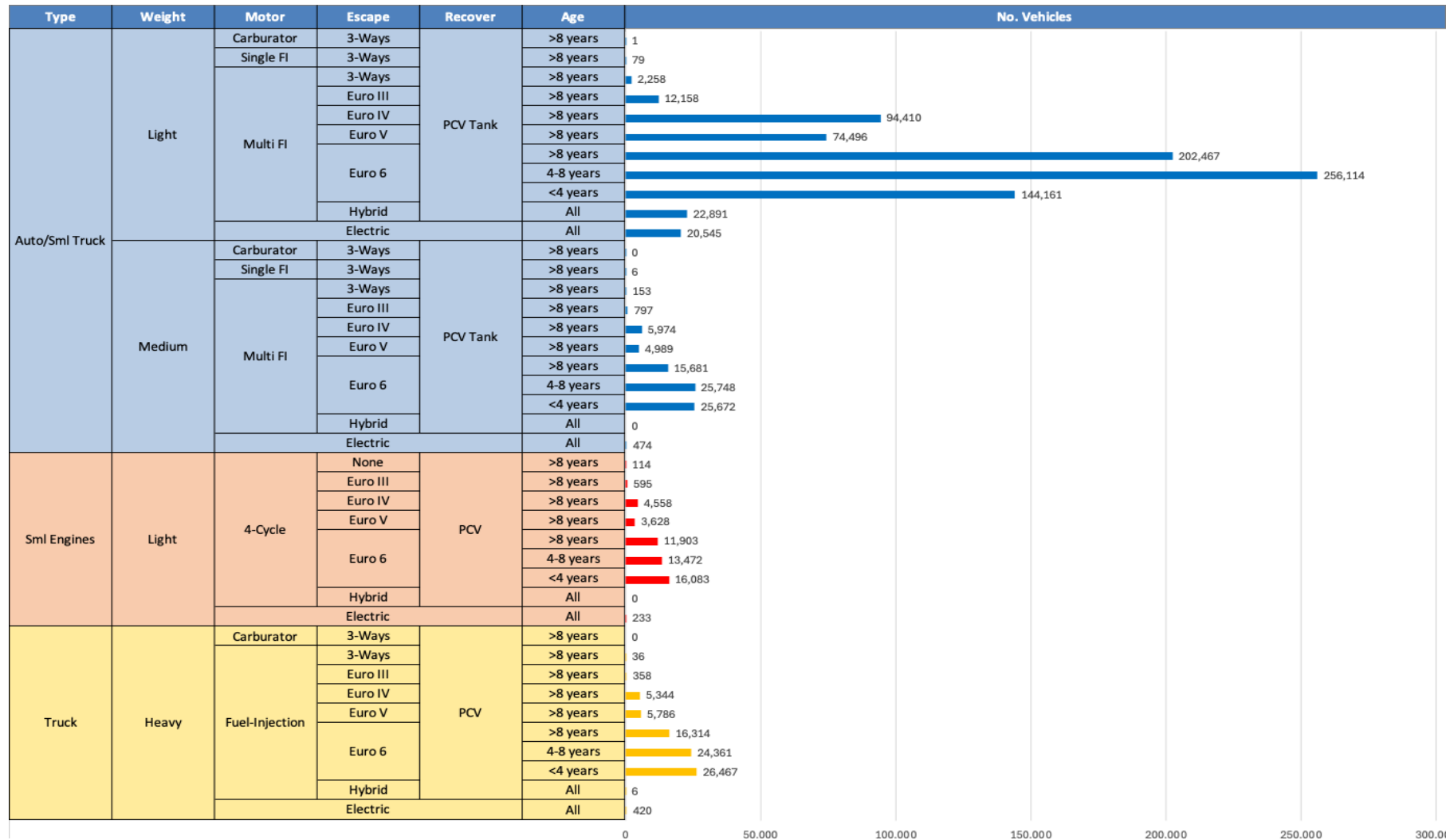
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Luxembourg

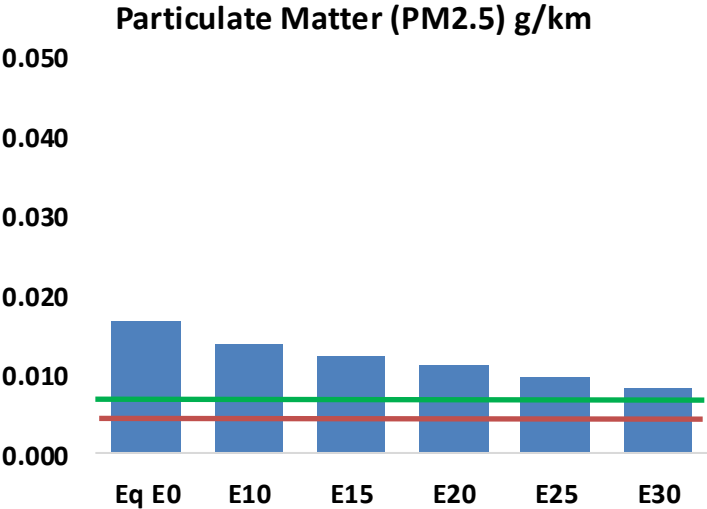
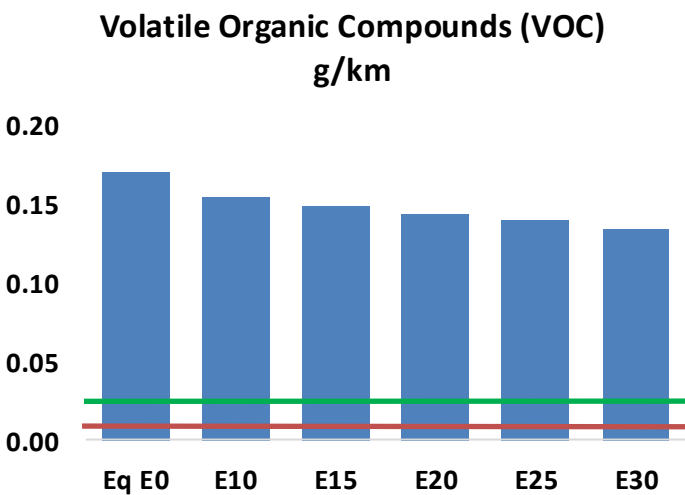
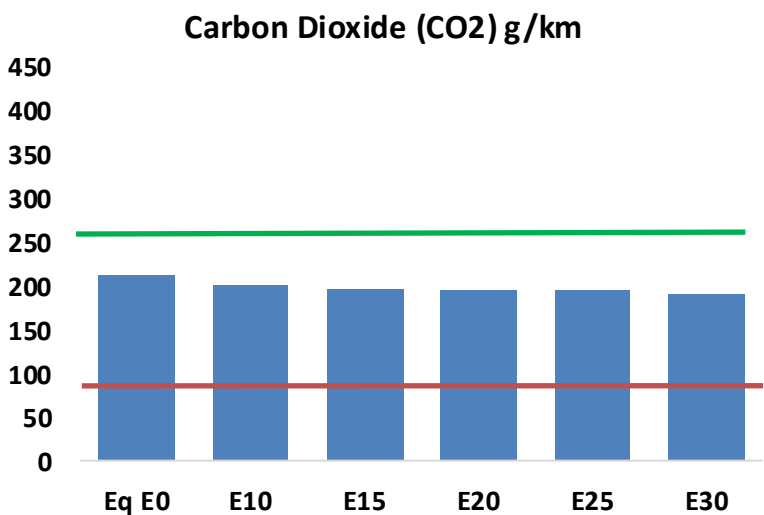
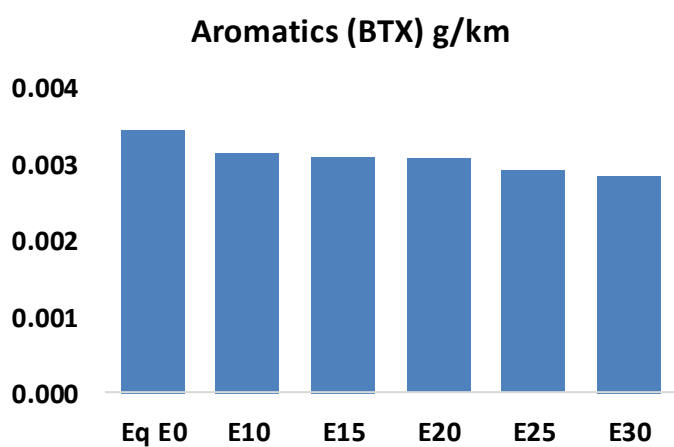
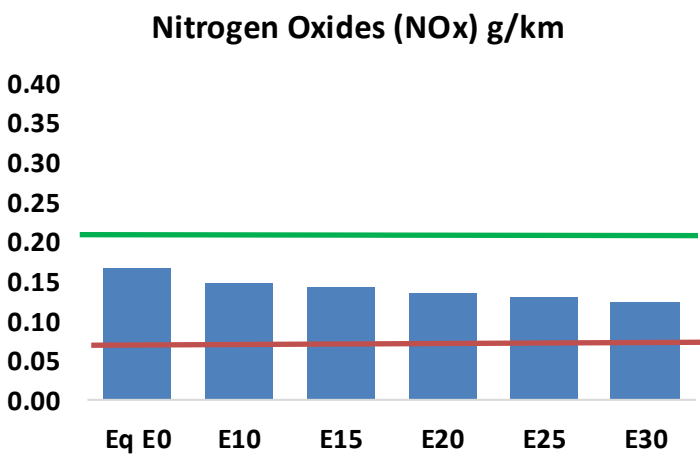
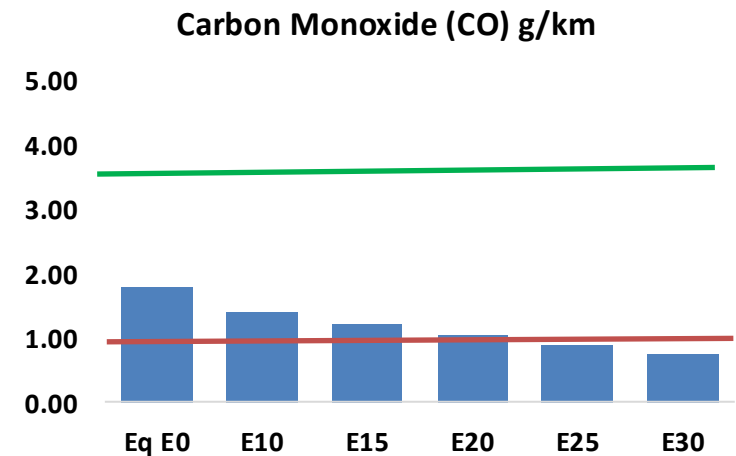
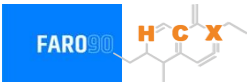


Vehicle Fleet: 1,038,755
Gasoline Vehicle Fleet: 694,397

Source: Eurostat, ACEA

Luxembourg – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Luxembourg – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	32,931	29,249	25,567	22,326	18,961	16,342	13,726	-22%	-42%
CO2	3,882,918	3,783,071	3,688,772	3,614,608	3,577,909	3,556,182	3,490,631	-5%	-8%
VOC	3,142	2,992	2,843	2,738	2,642	2,570	2,465	-10%	-16%
NOx	3,081	2,845	2,689	2,613	2,487	2,389	2,273	-13%	-19%
SOx	44	41	37	34	32	29	26	-15%	-28%
NH3	1,305	1,292	1,279	1,285	1,299	1,309	1,318	-2%	0%
N2O	48	47	47	47	48	49	49	-1%	2%
CH4	747	747	747	758	775	789	802	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	31	29	28	27	27	26	25	-8%	-14%
Acetaldehyde	55	64	93	142	194	221	261	68%	249%
Formaldehyde	162	157	184	216	226	250	272	13%	39%
Benzene	63	61	58	57	56	54	52	-9%	-11%
PM2.5	231	210	189	170	152	133	113	-18%	-34%
PM10	76	69	62	56	50	44	37	-18%	-34%

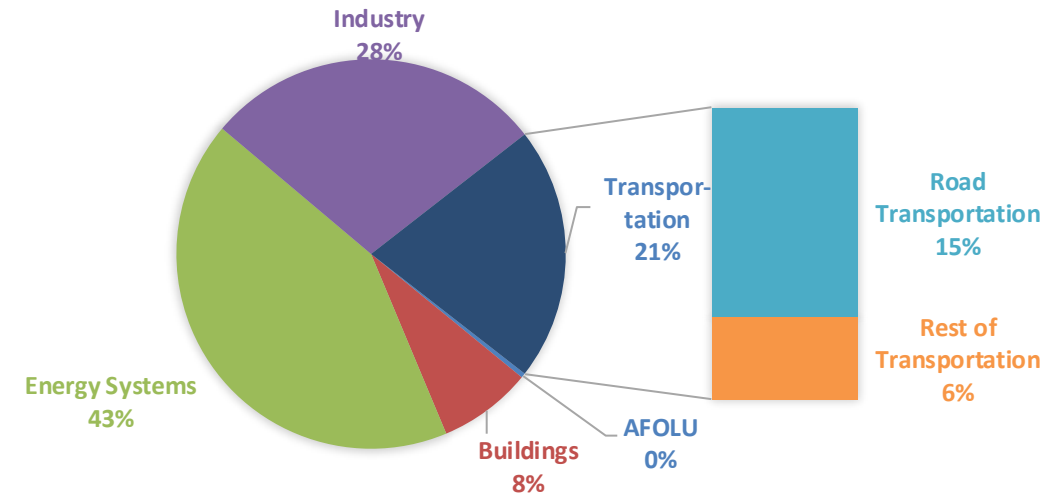
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

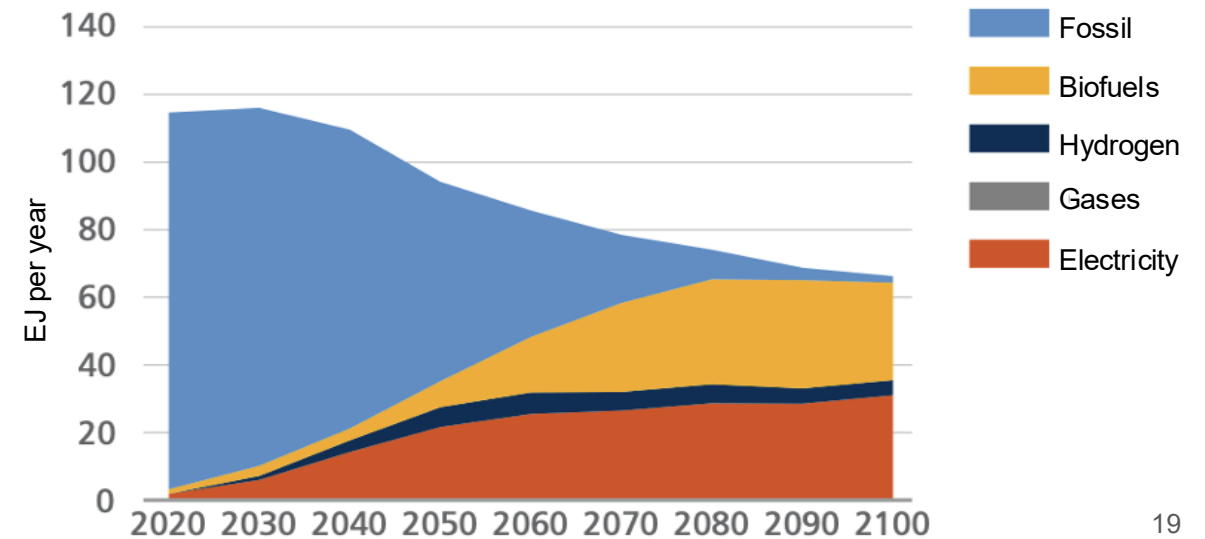
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

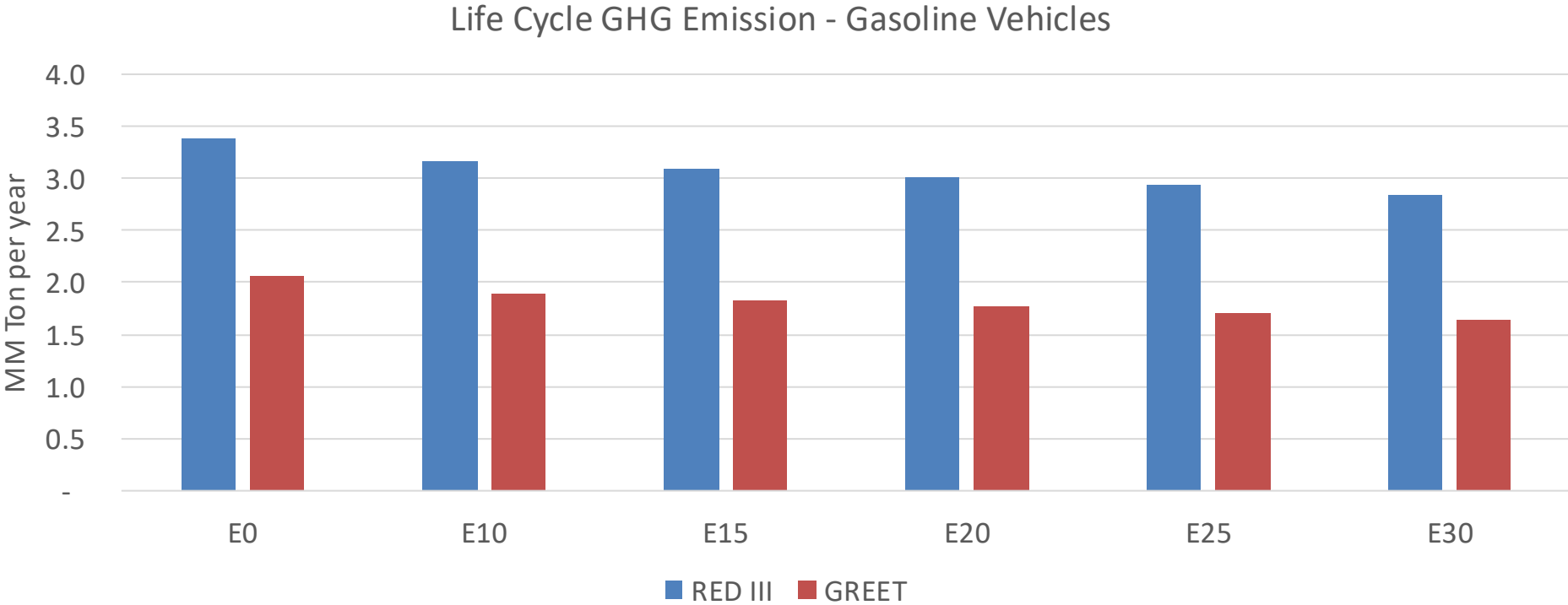
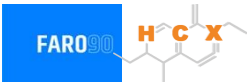
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Luxembourg – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	9.0
Target @ 2035 (MMT/year)	5.1
Delta	3.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.4%	11.6%	14.5%	17.1%	20.5%
RED II/III	0.0%	6.4%	8.9%	11.2%	13.3%	15.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.4%	6.1%	7.6%	8.9%	10.8%
RED II/III	0.0%	5.5%	7.7%	9.7%	11.4%	13.7%